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Aim: To understand DevOps, principles, practices and DevOps roles and responsibilities.

1. Introduction to DevOps

DevOps is a combination of **development (Dev)** and **operations (Ops)** that aims to improve collaboration, automation, and software delivery speed.

- DevOps helps organizations **deliver applications at high velocity**
- It improves **communication, collaboration, and integration**

◇ Problem Before DevOps

- Slow deployment (months instead of days)
- Conflicts between developers and operations
- Bugs found late in testing
- Inconsistent environments (dev, test, production)

2. DevOps Principles (CALMS Framework)

The **CALMS model** defines core DevOps principles:

1. Culture

- Focus on collaboration between teams
- Shared responsibility for success

2. Automation

- Reduce manual work
- Automate testing, deployment, and infrastructure

3. Lean

- Eliminate waste
- Improve efficiency and workflow

4. Measurement

- Track performance metrics like:
  - Deployment frequency
  - Failure rate
  - Recovery time

5. Sharing

- Share knowledge across teams
- Improve communication and transparency

3. DevOps Practices

1. Continuous Development

- Planning, coding, and version control
- Frequent updates to codebase

2. Continuous Integration (CI)

- Code is merged frequently
- Automated builds and testing are triggered
- Benefit: Detect bugs early

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3. Continuous Testing

- Automated testing done repeatedly

  Ensures continuous quality improvement

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4. Continuous Delivery & Deployment

- **Delivery**: Code is ready for release
- **Deployment**: Code is automatically deployed

  Faster and reliable releases

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5. Infrastructure as Code (IaC)

- Infrastructure managed using code
- Stored in version control

  Ensures consistency across environments

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6. Configuration Management

- Maintains system consistency

  👉Avoids environment mismatch issues

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7. Microservices Architecture

- Application divided into smaller services
  Improves scalability and flexibility

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8. Containerization

- Use of containers (e.g., Docker)
  Ensures application runs consistently everywhere

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9. Continuous Monitoring

- Tracks system performance and health
  Helps detect issues early

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10. Cloud-Based DevOps

- Use of cloud platforms for automation and scaling
  Improves flexibility and resource usage

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4. DevOps Engineer Roles & Responsibilities

A DevOps Engineer acts as a **bridge between development and operations teams**.

◆ **Key Responsibilities:**

1. CI/CD Pipeline Management

- Design and maintain automated pipelines
- Ensure smooth integration and deployment

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2. Automation

- Automate build, testing, and deployment processes
- Reduce manual intervention

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3. Infrastructure Management

- Use Infrastructure as Code tools
- Manage servers and cloud resources

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4. Monitoring & Logging

- Monitor system performance
- Analyze logs and resolve issues

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5. Collaboration

- Work with developers, QA, and operations teams
- Improve communication

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6. Security (DevSecOps)

- Implement security practices in pipelines
- Identify vulnerabilities early

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7. Troubleshooting

- Identify and fix production issues
- Ensure system reliability

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8. Version Control Management

- Maintain code repositories (Git)
- Ensure proper versioning

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